# GAITHO KARUOYA

SOFTWARE DEVELOPER & DATA ANALYST

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## PROFILE

I build production data systems end to end, from database schema to the interface people utilise — mainly in PostgreSQL, R Shiny and Linux deployment. I am currently the sole developer on two projects at ANOVA101: PollCentral and OSCE Training.

## CORE COMPETENCIES

- Relational database architecture
- Production web application development
- Statistical & psychometric analysis
- Geospatial data & mapping
- Survey & electoral data systems
- Deployment & system maintenance
- Technical troubleshooting & IT support

## TECHNICAL STACK

| | |
|---|---|
| **DATABASES** | PostgreSQL, SQL, DuckDB |
| **R** | Shiny, Tidyverse, DBI, pool, sf, Leaflet, echarts4r, reactable, pdftools |
| **WEB & LANGUAGES** | JavaScript, HTML, CSS, Python, Java, C, C++ |
| **INFRASTRUCTURE** | Docker, Linux server admin, cron, Cloudflare R2, shinyapps.io |
| **ANALYSIS** | Bayesian methods (MCMCpack), mixed-effects models (lme4), psychometrics (psych), MATLAB, SPSS |

## EXPERIENCE

### Statistical Programmer & Data Analyst — ANOVA101

Mar 2025 – Present

Sole developer of two production web applications.

**Common to both**

- **Stack:** modular R Shiny on PostgreSQL, with business rules enforced in the database through unique keys and check constraints, so invalid records are rejected even if the application has a bug.
- **Security:** passwords hashed with scrypt plus a server-side pepper; media served through signed, expiring links.

**PollCentral** — *Political Polling Platform*

- **Database:** lambda-style aggregation — triggers keep live per-ward totals for the open wave, a cron job rebuilds each closed wave from the raw votes into cache tables, and queries merge the two.
- **Polling waves:** daily voting windows scheduled to election day on 10 August 2027, with an exclusion constraint against overlapping waves and one vote per race per wave.
- **ETL:** parsed IEBC voter-statistics PDFs into 47 counties, 290 constituencies, 1,450 wards and 27,273 registration centres, recovering 39 missing centres from 46,035 polling stations; compiled 1,978 candidacies.
- **Maps & analytics:** Leaflet maps coloured by leading party, built from OCHA and Red Cross shapefiles; win probabilities from a Dirichlet posterior; results filtered by race, wave and location down to a single centre.
- **Accounts & interface:** registration tied to a voter's registration centre, email verification codes and 30-day persistent login; mobile-first in English and Swahili, with custom CSS and JavaScript.

Live site: **pollcentral.net** · Docker on a Hetzner Linux server

**OSCE Training** — *Clinical Examiner Training & Calibration Platform*

- **Roles:** trainee examiners rate recorded exam stations and compare their marks with reference examiners'; admins manage exams, stations, media and reference data.
- **Database:** check constraints reject ratings whose domain scores don't sum to the overall; transactional import of ~3,200 reference marks with a dry-run preview.
- **Analytics:** standard setting by Borderline Regression, Contrasting Groups and Borderline Group; Cronbach's alpha with SEM; an lme4 mixed-effects model of examiner severity.

Hosted on shinyapps.io · Videos on Cloudflare R2 behind a hand-written AWS SigV4 signer

### IT Consultant — Self-Employed
Dec 2023 – Mar 2025

- IT support and technical consultancy for private and institutional clients, mostly diagnosing hardware, software and system problems remotely.
- Statistical work for client research projects in MATLAB and SPSS.

### Data Clerk — Metrofile Records Management Kenya Ltd
Feb 2021 – Feb 2022

- Digitised client records at volume — scanning, keying in and checking the data as it went into client databases.
- Handled on-demand retrieval requests on client sites, working with supervisors to deadlines set per site.

### Software Developer Intern — Liden Company Limited
Jun – Aug 2020

- Debugged and tested application code alongside the senior developers, and wrote technical documentation.
- Turned requirements into working interface components.

### Software Developer Intern — Southwell Solutions Africa – VasPro
May – Oct 2019

- Worked on database design and API integration, mapping how the existing system worked from its codebase.
- Ran the weekly randomised draw for a client's promotion campaign.

## EDUCATION

### BSc Information Technology
2016 – 2020

*Multimedia University of Kenya · Second Class Honours, Upper Division*

### Kenya Certificate of Secondary Education
2012 – 2015

*Nyeri High School · Mean grade B+ (72 points)*

## REFEREES

Available on request.